notation, respectively. Notice that we are representing frequency by ω, the *natural frequency* in *radians per second*. If it makes you more comfortable, you can replace each ω with $2\pi f$ to make the expressions in hertz. However, most DSP mathematics is written using the shorter notation, and you should become familiar with it. Since it requires two parameters to represent a single sinusoid (i.e., *A* & *B*, or *M* & ϕ), the use of complex numbers to represent these important waveforms is a natural. Using substitution, the change from the conventional sinusoid representation to a complex number is straightforward. In rectangular form:

$$A\cos(\omega t) + B\sin(\omega t) \quad \rightleftarrows \quad a + jb$$

(conventional representation) (complex number)

where $A \rightleftarrows a$, and $B \rightleftarrows -b$. Put in words, the amplitude of the cosine wave becomes the real part of the complex number, while the *negative* of the sine wave's amplitude becomes the imaginary part. It is important to understand that this is *not* an equation, but merely a way of letting a complex number *represent* a sinusoid. This substitution also can be applied in polar form:

$$M\cos(\omega t + \phi) \quad \rightleftarrows \quad Me^{j\theta}$$

(conventional representation) (complex number)

where $M \rightleftarrows M$, and $\theta \rightleftarrows -\phi$. In words, the polar notation substitution leaves the magnitude the same, but changes the sign of the phase angle.

Why change the sign of the imaginary part & phase angle? This is to make the substitution appear in the same form as the *complex Fourier transform* described in the next chapter. The *substitution* techniques of this chapter gain nothing from this sign change, but it is almost always done to keep things consistent with the more advanced methods.

Using complex numbers to represent sine and cosine waves is a common technique in electrical circuit analysis and DSP. This is because many (but not all) of the rules and laws governing complex numbers are the same as those governing sinusoids. In other words, we can represent the sine and cosine waves with complex numbers, manipulate the numbers in various ways, and have the resulting answer match the way the sinusoids behave.

However, we must be careful to use only those mathematical operations that mimic the physical problem being represented (sinusoids in this case). For example, suppose we use the complex variables, *A* and *B*, to represent two sinusoids of the same frequency, but with different amplitudes and phase shifts. When the two complex numbers are *added*, a third complex number is produced. Likewise, a third sinusoid is created when the two sinusoids are